



XAPP587 (v1.0.1) May 31, 2012

BPI Fast Configuration and iMPACT Flash Programming with 7 Series FPGAs

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Summary

This application note describes the 7 series FPGA Byte-wide Peripheral Interface (BPI) configuration mode with synchronous read and the External Master Configuration Clock (EMCCLK) options. Because the BPI configuration mode with synchronous read and EMCCLK provides the fastest configuration time from a direct FPGA interface, it is referred to as *BPI Fast Configuration* in this document. The BPI Fast Configuration enables high-capacity non-volatile storage and decreases the configuration time to less than 8% of the legacy BPI configuration with the asynchronous read. The *7 Series FPGAs Configuration User Guide* [Ref 1] contains details on FPGA configuration. Familiarity with this document is assumed.

The hardware setup, file generation, and the tool flow to indirectly program a parallel NOR flash through the 7 series FPGA are described. The Virtex®-7 FPGA and 28F00AG18F parallel NOR flash on the VC707 evaluation board are used to demonstrate the flow with the ISE® Design Suite 14.1.

Introduction

Figure 1 shows the two flows that are described in this document:

- 7 series BPI Fast Configuration process
- iMPACT indirect flash programming via the FPGA

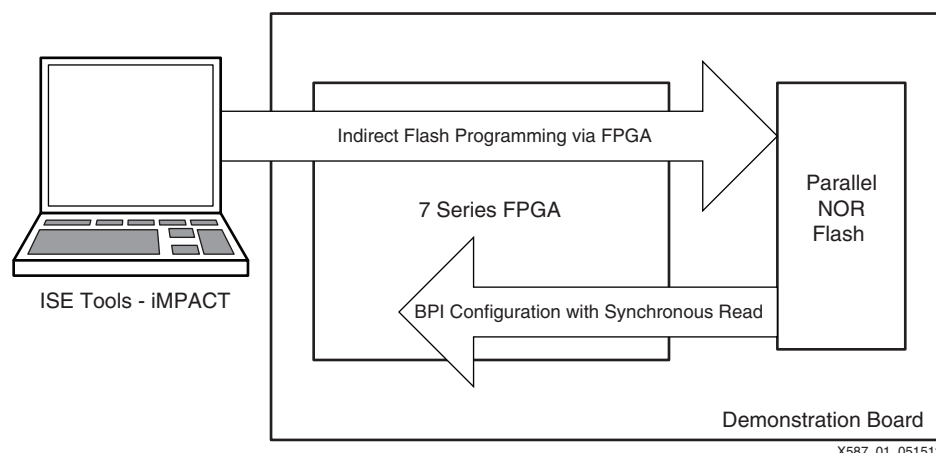


Figure 1: BPI Fast Configuration and iMPACT Indirect Flash Programming Flows

7 series FPGAs require a configuration bitstream to be delivered at power-up because they are CMOS-configurable latch-based devices. Parallel NOR flash is a popular option for storing and delivering the bitstream because the wide x16 data bus provides faster configuration over SPI flash alternatives. Specifications such as PCI Express® with strict power-up requirements need fast configuration like those achieved with the BPI Fast Configuration. In addition to the Fast Configuration, parallel NOR flash offers larger capacity than SPI flash. Systems that use parallel NOR flash memory for random-access, non-volatile application data storage can also benefit from consolidating the configuration storage into a single memory device.

Xilinx also provides the ability to indirectly program parallel NOR flash in-system using the existing configuration connections between the parallel NOR flash and the FPGA. The iMPACT programming tool uses JTAG to configure the FPGA with a pre-made bitstream. This bitstream enables a path between the programming cable and the flash. In a lab environment, this solution provides the flexibility to easily program new configuration bitstreams into the parallel NOR flash without needing to remove the device from the board.

This application note is divided into the following key sections:

- [BPI Configuration Basics, page 2](#)
- [Parallel NOR Flash Programming Basics, page 2](#)
- [BPI Fast Configuration Implementation, page 3](#)
- [BPI Fast Configuration File Generation and Flash Programming, page 10](#)
- [Checklist and Debug, page 20](#)

BPI Configuration Basics

Configuration is the process of downloading configuration data into an FPGA using an external source such as a flash device, CPLD, or microprocessor. In the BPI configuration mode, the FPGA supports a direct connection to the address, x16 data bus, and control signals of a parallel NOR flash for extracting a stored bitstream. The 7 series FPGA BPI configuration interface supports two flash read options: asynchronous or synchronous. The BPI configuration mode supports the configuration data download by using either the FPGA CCLK (internally-generated configuration clock) or the EMCCLK option.

Synchronous Read and EMCCLK Advantages

BPI Fast Configuration uses the synchronous read and EMCCLK options. Synchronous read can burst configuration data at frequencies up to 100 MHz after a valid starting address is given. This significantly decreases the configuration time from what can be achieved with the asynchronous read mode. In the legacy asynchronous read mode, the FPGA must increment the address at a frequency less than 10 MHz to support the supported flash read access specifications.

The configuration times can be further decreased by using the EMCCLK option in conjunction with the synchronous read. EMCCLK uses external oscillators that are capable of faster and more precise clock frequency options than provided with the FPGA's internal CCLK. This enables more predictable configuration times. For example, if the maximum clock rate for a synchronous read setup is 100 MHz, an external clock of 99 MHz can be used for EMCCLK if the tolerance is 1%. This is an enhancement over the maximum 66 MHz CCLK rate for the same setup. The CCLK maximum is a lower target because of the tolerance $F_{MCCKTOL}$. For example, at a typical $F_{MCCKTOL}$ of $\pm 50\%$ the wide range of 33–99 MHz must be taken into account. [Configuration Times, page 9](#) describes how to estimate the configuration time.

Parallel NOR Flash Programming Basics

Parallel NOR flash memory devices have a single interface for programming and three primary methods to deliver the data to this interface:

- Third-party programmers (off-board programming)
- In-system programming (ISP) with an embedded processor
- Indirect ISP using JTAG or a custom solution

Key factors to consider in selecting the best programming solution for an application are:

- Prototyping or high-volume production current phase
- ISP or off-board programming required
- Direct or indirect programming connectivity available

For example, an off-board third-party programmer such as BP Microsystems should be considered if the most critical factor is to decrease flash programming times for a high-volume production application. Off-board programming can often deliver faster programming times because they can limit overhead by interfacing directly to the flash. This solution can also make use of the enhanced programming higher voltage option.

An application in the low-volume prototyping phase requiring the flash to be programmed on-board would benefit from the iMPACT indirect ISP programming tool. The iMPACT tool can accommodate the testing of multiple design iterations and is extremely useful for debugging in a lab environment, but it is not intended for high volume production programming.

BPI Fast Configuration Implementation

7 series FPGAs have a configuration interface to third-party parallel NOR flash. The 7 series BPI configuration mode supports asynchronous read as in earlier FPGA families, but to address changing system requirements, new BPI configuration mode features have been added. The 7 series FPGAs are the first Xilinx FPGAs with the built-in ability to synchronously read from select parallel NOR flash memory for configuration. In addition, an EMCCLK option can be enabled to provide a highly accurate configuration clock. These two new features used together enable significantly faster configuration times than in past generations. The sections that follow discuss the criteria for selecting a flash device and describe the interface and BPI Fast Configuration sequence.

Selecting Parallel NOR Flash

Several factors should be considered when selecting the parallel NOR flash device for a configuration source, such as the storage capacity required by the application, the package type to meet board space requirements, the data bus width for configuration time, and the flash I/O voltage range. See the *7 Series FPGAs Configuration User Guide* [Ref 1] for information on the bitstream size to determine the minimum flash density required.

Because the flash must be placed into synchronous read mode with a write command sequence, only the Micron G18 and P30 flash families are supported. Within the Micron G18 family, only the Non-MUX interface devices are supported. This application note focuses on the G18 family because it is capable of running twice the frequency of the P30 family. Table 1 shows the supported devices from the G18 family.

Table 1: BPI Fast Configuration Parallel NOR Flash

Flash Family	Density Range	V _{CC} /V _{CCQ} (I/O)	iMPACT Indirect Programming
Micron G18	128 Mb–1 Gb	1.8V/1.8V	See iMPACT manual [Ref 2] for release details.

Voltage Compatibility

The parallel NOR flash I/O must support a voltage that is compatible with the connected FPGA pins. The dedicated configuration pins in bank 0 and the dual-purpose pins in banks 14 and 15 are used by the BPI Fast Configuration interface. These pins need to receive the same V_{CCO} voltage supply to ensure a consistent I/O voltage for the BPI configuration interface. In addition, the JTAG pins are in bank 0 and follow the requirements for this bank. The configuration bank voltage select pin (CFGBVS) is set High in order to set the configuration-related I/O for 3.3V/2.5V operation, or set Low to set the configuration pins for 1.8V operation. This application note focuses on the Virtex-7 FPGA whose I/O are limited to 1.8V or lower standards on banks 14 and 15. In the Virtex-7 FPGA, the CFGBVS pin is set to Low to support BPI Fast Configuration. For additional details on CFGBVS, refer to the *7 Series FPGAs Configuration User Guide* [Ref 1].

BPI Fast Configuration Interface

Figure 2 shows associated signals for the BPI Fast Configuration interface. Refer to [BPI Fast Configuration Sequence, page 8](#) for details on the configuration process steps for this mode.

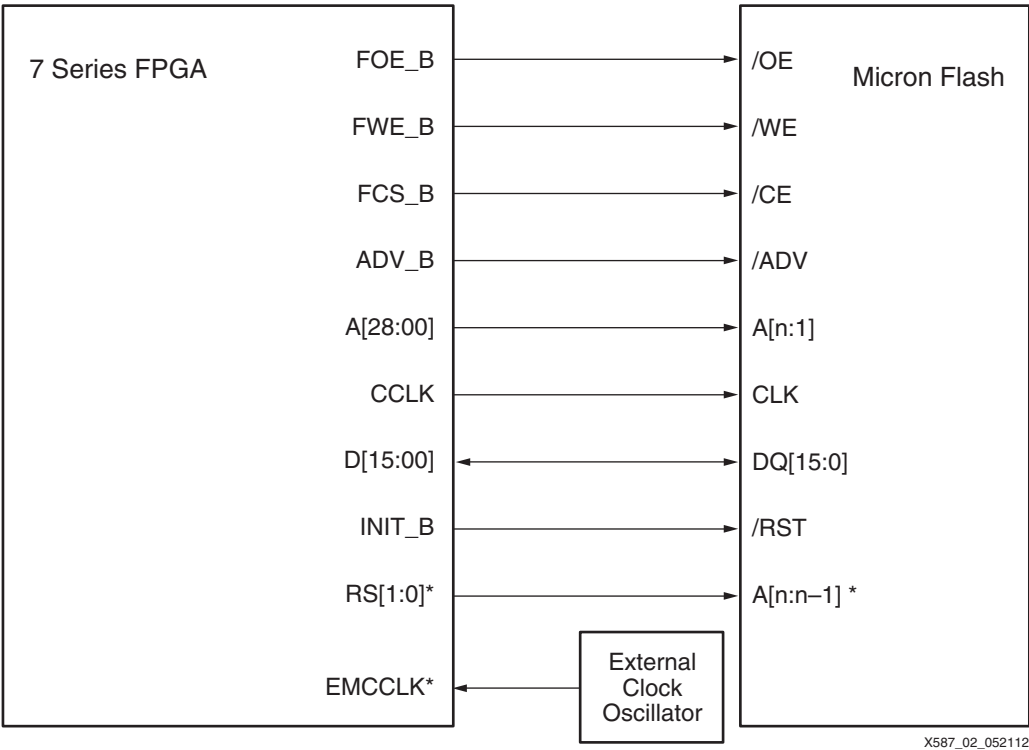


Figure 2: BPI Fast Configuration Interface

Note: The RS[1:0] and EMCCLK signals are optional connections for the BPI configuration mode highlighted in this application note.

[Table 2](#) describes the BPI Fast Configuration interface signals and their function in the configuration process. The connections to the parallel NOR flash memory are also provided.

Table 2: 7 Series FPGA BPI Configuration Signal Descriptions

7 Series FPGA Pin Name	Type	Dedicated or Dual-Purpose	Description	Micron Flash Signal Connection ⁽¹⁾
CCLK	Input or Output	Dedicated	Configuration clock output. This pin is the initial configuration clock source for all configuration modes except JTAG. CCLK is an output in master BPI configuration mode. During the BPI flash asynchronous read mode, CCLK does not directly clock the parallel NOR flash but is used internally by the FPGA to generate the address and sample read data. During the BPI flash synchronous read mode, CCLK must be directly connected to the parallel NOR flash to clock the data out sequentially.	CLK (Clock)
DONE	Bidirectional, Open-drain, or Active	Dedicated	Active-High signal indicating configuration is complete: 0 = FPGA not configured 1 = FPGA configured	NC

Table 2: 7 Series FPGA BPI Configuration Signal Descriptions (Cont'd)

7 Series FPGA Pin Name	Type	Dedicated or Dual-Purpose	Description	Micron Flash Signal Connection ⁽¹⁾
INIT_B	Bidirectional	Dedicated	From power-on reset or PROGRAM_B reset, INIT_B is driven Low, indicating that the FPGA is initializing (clearing) its configuration memory. Before the mode pins are sampled, INIT_B is an open-drain, active-Low input and can be held Low to delay configuration. After the mode pins are sampled, the INIT_B output indicates whether a CRC error occurred during configuration or a readback CRC error occurred after configuration (when enabled): 0 = CRC or IDCODE error (DONE is Low) or readback CRC error (DONE is High and readback CRC is enabled) 1 = No CRC error, initialization is complete (needs an external pull-up resistor to VCCO_0)	/RST (Reset)
M[2:0]	Input	Dedicated	The mode pins determine the configuration mode. M[2:0]=010 for Master BPI mode.	NC
PROGRAM_B	Input	Dedicated	Active-Low asynchronous full-chip reset.	NC
A[28:00]	Output	Dual-purpose	Address output. Signals are used to increment the flash address and in synchronous read mode, these signals send the flash read configuration register content that set the flash in synchronous mode.	A[n:1] (Address inputs)
ADV_B	Output	Dual-purpose	Active-Low address valid output. Required for the synchronous read option in Master BPI Configuration mode. This signal is needed to tell when an address is valid in synchronous read mode. The signal also needs to be driven Low for asynchronous read mode.	/ADV (Address valid)
CSO_B	3-state output	Dual-purpose	Parallel daisy chain Active-Low chip select output during configuration. Otherwise, this output is 3-stated. Not used in single FPGA applications.	NC
D[15:00]	Bidirectional	Dual-purpose	This x16 data bus is sampled by the rising edge of the FPGA CCLK. Data is read from the flash on this bus and commands to write to the flash read configuration register are sent on this bus when a synchronous read command is seen by the configuration controller.	DQ[15:0] (Data)
EMCCLK	Input	Dual-purpose	External Master Configuration Clock input. This option is used for the BPI Fast Configuration mode. An external clock is supplied on this input and the FPGA configuration controller switches over to use this clock instead of the CCLK (internal configuration clock) after the EMCCLK command is read from the bitstream header. The EMCCLK enables more predictable configuration times because the clock tolerance is determined by the external oscillator selected.	NC
FCS_B	Output	Dual-purpose	Active-Low flash chip select output. This output is actively toggled during configuration.	/CE (Chip Enable)

Table 2: 7 Series FPGA BPI Configuration Signal Descriptions (Cont'd)

7 Series FPGA Pin Name	Type	Dedicated or Dual-Purpose	Description	Micron Flash Signal Connection ⁽¹⁾
FOE_B	Output	Dual-purpose	Active-Low flash output enable. This output is actively toggled during configuration.	/OE (Output Enable)
FWE_B	Output	Dual-purpose	Active-Low flash write enable. This output is actively toggled during configuration.	/WE (Write Enable)
PUDC_B	Output	Dual-purpose	Controls I/O (except bank 0 dedicated I/O) pull-up resistors during configuration. This pin must be externally terminated: 0 = Pull-up during configuration 1 = 3-state during configuration	NC
RS[1:0]	Output	Dual-purpose	Revision Select pins. Used for multi-bitstream applications to easily select between revisions and provide fallback capability. RS[1:0] are actively driven Low to load the fallback bitstream when a configuration error is detected. See the <i>7 Series FPGAs Configuration User Guide</i> [Ref 1] for additional details on revision management.	A[n:n-1] (Upper address bits) ⁽²⁾

Notes:

1. Refer to Micron's G18 flash data sheet [Ref 3] for flash signal descriptions and details.
2. The use of the revision select pins (RS[1:0]) is an optional feature used in cases where bitstream revision management is required.

Figure 3 shows the BPI Fast Configuration example implementation on the VC707 evaluation board. The required signal connections to the parallel NOR flash are given in the figure. Signal connections for the JTAG port connections to support the iMPACT indirect flash programming are included.

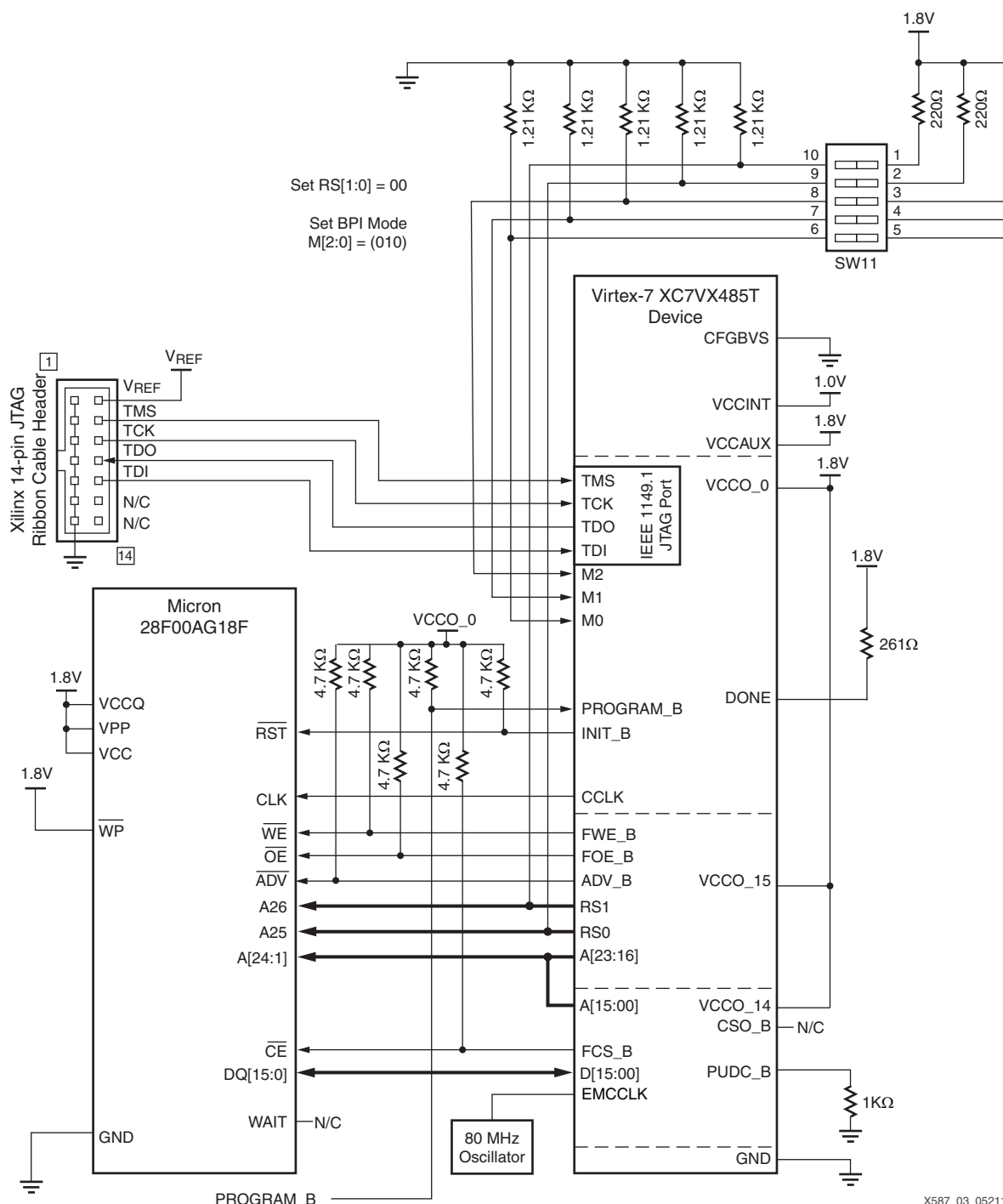


Figure 3: VC707 Sample BPI Fast Configuration Setup

Notes relevant to [Figure 3](#):

1. The 7 series FPGA JTAG signals TCK, TMS, TDI, and TDO are not required for the BPI Fast Configuration interface, but these signals are necessary for the iMPACT indirect flash programming solution. The JTAG interface is also a popular debug interface used on many application setups. Refer to [BPI Fast Configuration File Generation and Flash Programming, page 10](#) for steps to program the flash indirectly.

2. The EMCCLK maximum frequency is dependent on the target flash and FPGA. See [Configuration Times, page 9](#) to determine the maximum frequency for a given setup. The VC707 board uses an 80 MHz clock source.
3. RS[1:0] are optional revision management pins. In applications that do not require revision control, the required address pins are connected from FPGA A[28:0] to the flash A[n:1] and RS[1:0] is a no connect.
4. The 7 series FPGA VCCO_0 supply must be compatible with the V_{CCQ} on the parallel flash.
5. The Micron 28F00AG18F has Write Protect (/WP) and Wait signals that should be tied appropriately to allow for indirect flash programming and configuration.

BPI Fast Configuration Sequence

The BPI Fast Configuration sequence described in this section occurs automatically when the bitstream has been generated for the synchronous read mode and the mode pins M[2:0] are set to the BPI configuration mode. Refer to [BPI Fast Configuration File Generation and Flash Programming, page 10](#) for the steps to generate the bitstream.

In the BPI configuration mode, the 7 series FPGA always begins reading from the parallel NOR flash in asynchronous read mode. The FPGA increments the address bus from a given start address, and the parallel NOR flash sends back the bitstream data. The 7 series FPGA reads the bitstream header to determine the flash read mode selected for reading the configuration data. When a synchronous command is read in the bitstream header, the FPGA configuration controller performs an asynchronous write to the Read Configuration register (RCR) of the connected flash to set the synchronous mode and latency bits. After the flash RCR has been successfully written, the FPGA controller then initiates synchronous read and reads the bitstream data contents. After configuration completes, the flash is left in synchronous read mode.

The timing waveform in [Figure 4](#) shows the three steps to initiate the BPI configuration synchronous read.

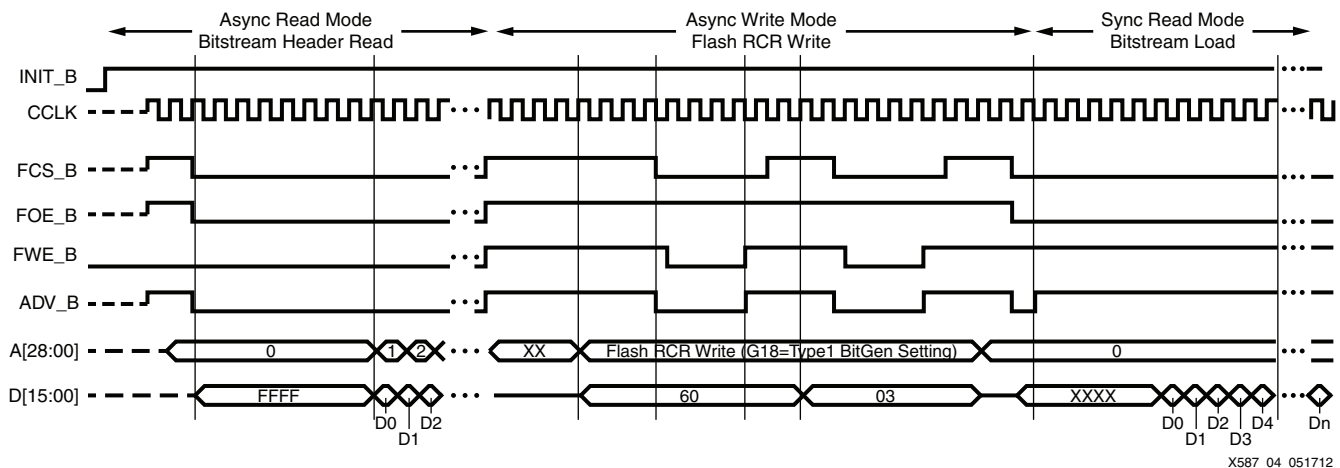


Figure 4: BPI Fast Configuration Timing Waveform

The first step is reading the bitstream in asynchronous mode to determine the target mode. After the INIT_B signal is released and the control signals FCS_B, FOE, and ADV_B are asserted with a valid address A[28:00] being incremented, data is captured from the flash on the data bus D[15:0]. The bitstream header is read to determine which read mode is targeted by the design.

In the second step, the FPGA enables the flash synchronous mode by writing to the read configuration register of the flash. To write the flash synchronous mode bit, the FPGA issues an

asynchronous write sequence. To perform the write operation, the FPGA asserts the FCS_B and FWE_B while the INIT_B and FOE_B are deasserted. During the write operation, the address and data are latched on the rising edge of FWE_B or FCS_B, whichever occurs first. The FPGA issues the flash configuration register write sequence of two write cycles. The first cycle has the Read Configuration register (RCR) data on A[16:01] and command 0x60 on the data bus. The second cycle has the RCR data on A[16:01] and the command 0x03 on the data bus. The RCR values are different for the Micron G18 and the P30 families and is determined by the BitGen option **-g BPI_sync_mode**. Refer to [Table 4](#) for options.

In the final step, the FPGA finally switches from the asynchronous read to synchronous read protocol and reinitiates the bitstream read data contents. This sequence is implemented by asserting the FCS_B and the FOE_B signals and having ADV_B asserted for one cycle with a valid address. The configuration data is then burst from the flash and read back by the FPGA.

It is important to understand that the flash is left in the same read mode that is used for configuration. For example, the flash is left in asynchronous read mode after the FPGA is configured in the asynchronous read mode. Alternatively, the flash is left in synchronous read mode after the FPGA is configured in the synchronous read mode.

Configuration Times

The BPI Fast Configuration setup in this application note uses EMCCLK. To calculate the maximum EMCCLK frequency, both the parallel NOR flash clock to out specification and the FPGA setup data sheet specifications are used. In addition, board delay needs to be considered. The maximum supported EMCCLK frequency (F_{EMCCK}) is specified in the 7 series FPGA data sheets [\[Ref 4\]](#) [\[Ref 5\]](#) [\[Ref 6\]](#) and must not be exceeded. An estimation for the maximum BPI Fast Configuration EMCCLK can be calculated with [Equation 1](#).

$$\text{Maximum Frequency} = \frac{1}{\text{Flash clock to out (TCHQV)} + \text{FPGA data setup (TBPIDCC)} + \text{Board delay}} \quad \text{Equation 1}$$

Note: With a 28F00AG18F flash clock-to-out specification of $T_{CHQV} = 5.5$ ns and a Virtex-7 XC7VX485T FPGA data setup of $T_{BPIDCC} = 4$ ns, the maximum EMCCLK would be capped by the data sheet specification F_{EMCCK} if board delay was negligible.

The BPI Fast Configuration time on the VC707 board is given compared to the BPI Configuration mode with asynchronous read and CCLK on the same board in [Table 3](#). The calculation used to estimate the configuration time is given in [Equation 2](#).

$$\text{Configuration time} = \frac{\text{bitstream size}}{\text{configuration clock frequency} \times \text{data bus width}} \quad \text{Equation 2}$$

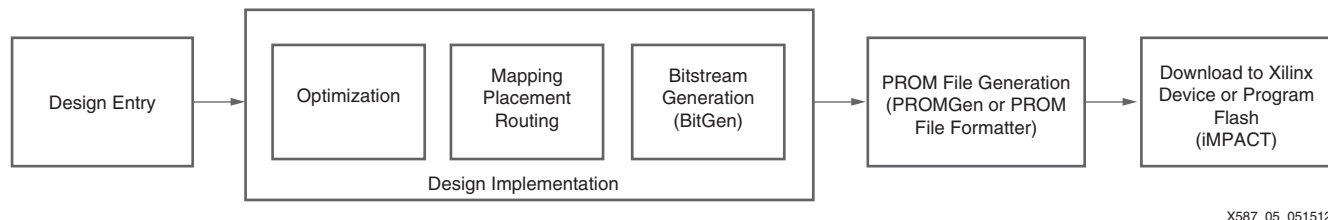
Table 3: Sample BPI Configuration Times for VC707 Board (XC7VX485T FPGA from 28F00AG18F Flash Memory)

Read Mode	Configuration Clock Source	Flash Data Width	Configuration Time for Bitstream (162,187,488 Bits)
Synchronous Read	EMCCLK (VC707 80 MHz oscillator)	x16	126 ms
Asynchronous Read	CCLK = 6 MHz	x16	1.7s

For the asynchronous read calculation, additional parameters must be considered such as the FPGA T_{BPICCO} + Flash T_{ACC} + FPGA data setup T_{BPIDCC} , which determines the maximum clock frequency. After the maximum clock frequency is known, the CCLK tolerance ($F_{MCCKTOL}$) must be considered when using the internal configuration clock CCLK instead of the EMCCLK. Although the 28F00AG18F flash is capable of running at slightly more than 9 MHz in asynchronous read mode, the $F_{MCCKTOL}$ limits the CCLK frequency setting to 6 MHz typical (or a range of 3 MHz–9 MHz).

BPI Fast Configuration File Generation and Flash Programming

The ISE design tools create 7 series FPGA bitstreams and flash programming files, and indirectly program parallel NOR flash devices. [Figure 5](#) highlights the steps (Design Entry, BitGen, PROMGen, and iMPACT) where special options are required to properly generate files for the BPI Fast Configuration mode and program the flash. Refer to the Virtex-7 FPGA VC707 Evaluation Board demonstrations for reference design examples using this flow and options at http://www.xilinx.com/support/documentation/vc707_14-1.htm.



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Figure 5: ISE Design Tools Overview

This section provides setup and tool flow examples for the following target:

- ISE tools, version 14.1
- VC707 board (Virtex-7 XC7VX485T FPGA with 28F00AG18F Parallel NOR flash and USB cable)

Design Entry Preparation for EMCCLK

Applications that use the BPI Configuration EMCCLK option must ensure that the I/O standard is defined on the EMCCLK dual-purpose pin. For example, a design for the Virtex-7 FPGA on the VC707 board would need to include this user constraint option to enable the 80 MHz EMCCLK:

```
NET "EMCCLK" LOC = AP37 | IOSTANDARD = LVCMOS18;
```

Preparing a Bitstream for BPI Configuration

This section provides the options needed to create a bitstream that can be used with the fastest BPI Configuration mode using the synchronous read. [Table 4](#) provides the recommended options for the VC707 board and descriptions of the features related to the BPI Configuration mode. Options not specified in the BitGen command use the defaults.

Table 4: BitGen Configuration Options for BPI Configuration Mode with Synchronous Read

BitGen Command Line	Description
-g StartupClk:Cclk	Option: StartupClk: Cclk UserClk JtagClk Default: Cclk Specifies the FPGA CCLK as the clock that drives the FPGA startup sequence. The Cclk option is required for successful BPI configuration.
-g Compress	Option: Compress Default: Off Specifies that FPGA bitstream file compression is enabled. This is an optional switch but recommended to reduce both indirect flash programming and FPGA configuration time. Compression can significantly reduce the file size, but it is nondeterministic and varies with the user design.

Table 4: BitGen Configuration Options for BPI Configuration Mode with Synchronous Read (Cont'd)

BitGen Command Line	Description
-g ConfigRate:3	Option: ConfigRate:3 6 9 12 16 22 26 33 40 50 66 Default: 3 Specifies a nominal CCLK frequency of 3 MHz. The FPGA CCLK controls the FPGA BPI configuration timing. This application note assumes that EMCCLK is utilized so the default is recommended and only used for the reading of the initial bitstream header. For designs that do not utilize the EMCCLK, the ConfigRate option can be increased to a frequency that meets the maximum clock frequency supported for the FPGA and flash taking into account the F_{MCKTOL} tolerance.
-g BPI_sync_mode:Type1	Option: BPI_sync_mode:Disable Enable Default: Disable By default, the 7 series FPGA flash supports asynchronous read mode with this option disabled. For synchronous read, one of these settings must be selected: Type1 = G18 Type2 = P30 For this application note, Type1 is required.
-g ExtMasterCclk_en:div-1	Option: ExtMasterCclk_en:Disable div-8 div-4 div-2 div-1 Default: Disable External master configuration clock used by the BPI Configuration mode when an accurate CCLK is required (vs. an internal master CCLK that has a wide F_{MCKTOL} tolerance). EMCCLK divided by 1 is the maximum frequency selection used when the div-1 option is set. The div-2 option divides the EMCCLK by 2, the div-4 option divides the EMCCLK by 4, and so on for div-8. The user should ensure that the flash data sheet can support the frequency selected.
-g BPI_1st_read_cycle:1	Option: BPI_1st_read_cycle:1 2 3 4 Default: 1 This option must be left at the default because this feature is not supported by the BPI Configuration mode synchronous read.
-g BPI_page_size:1	Option: BPI_page_size:1 4 8 Default: 1 This option must be left at the default because this feature is not supported by the BPI Configuration mode synchronous read.

BitGen Command Line Example

The bitstream generation tool BitGen can be easily run from a command line. An example for the VC707 target with the BPI Configuration synchronous mode is shown here:

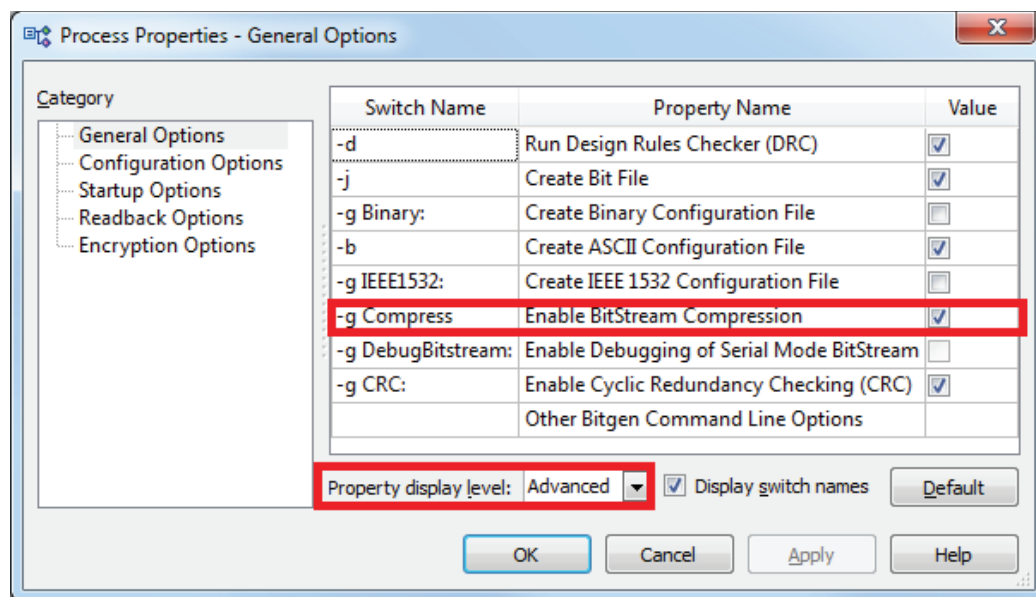
```
bitgen.exe -w -g StartUpClk:CCLK -g Compress -g ConfigRate:3 -g
BPI_sync_mode:Type1 -g ExtMasterCclk_en:div-1 -g BPI_page_size:1 -g
BPI_1st_read_cycle:1 design.ncd design.bit
```

Note: The VC707 board has an oscillator connected to the EMCCLK FPGA I/O of 80 MHz so the div-1 option can be used without exceeding the flash or FPGA maximum synchronous frequency.

BitGen Graphical User Interface Example

The bitstream can also be generated from the ISE tools Project Navigator flow (Figure 6). In the Project Navigator, the defaults are used together with these options:

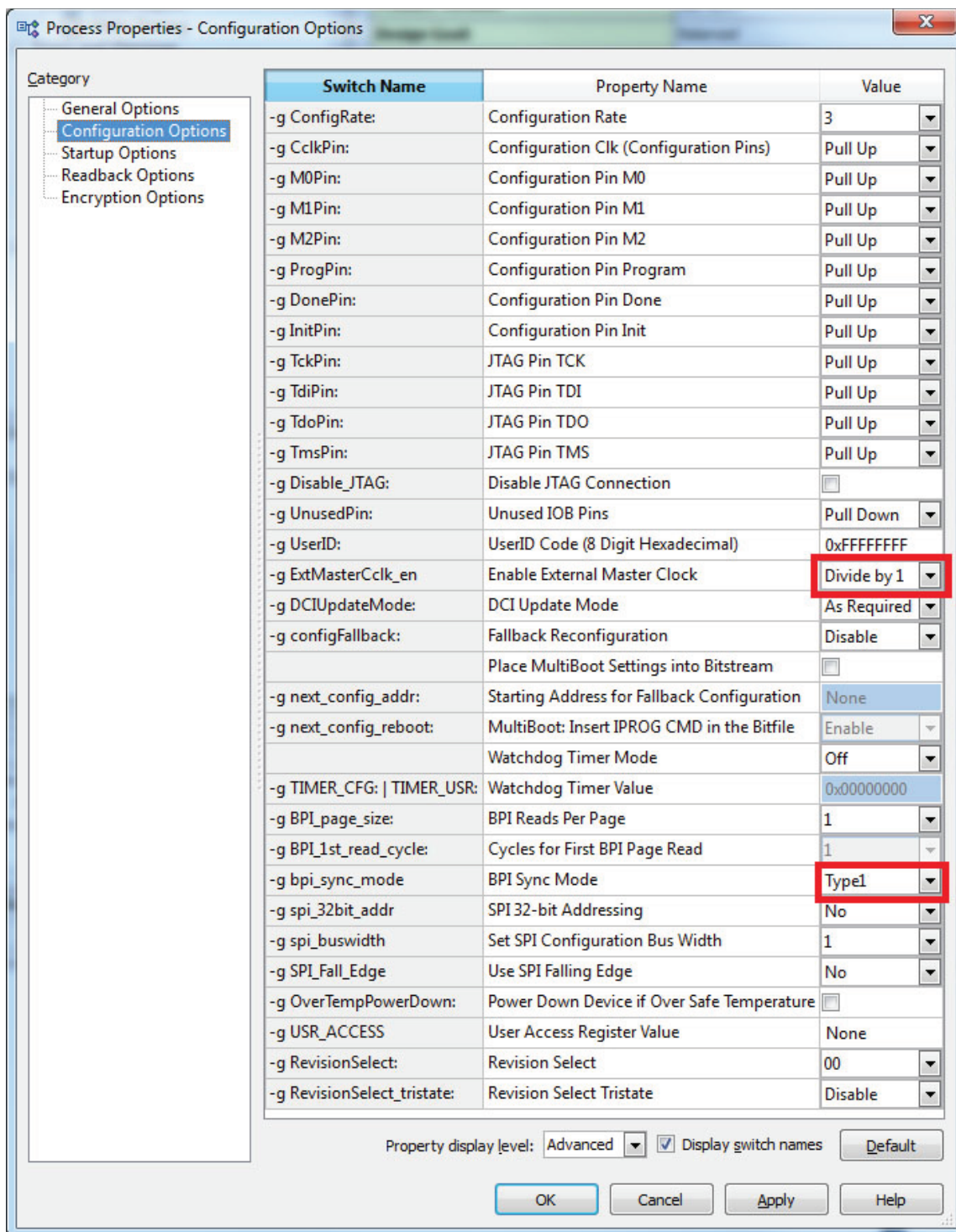
- Under the Generate Programming File process, right click to get the process properties:
 - For the Property Display Level, select **Advanced** to show all possible options.
 - Under General Options, the **-g Compress** switch is optional, but it is selected because it helps to reduce the programming and configuration time.



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Figure 6: General Options for BPI Fast Configuration Mode Bitstream Generation

- Under the Configuration Options category, ensure that the options shown in Figure 7 are selected to generate a bitstream for BPI configuration with synchronous read capability using the external master configuration clock.



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Figure 7: Configuration Options for BPI Fast Configuration Mode Bitstream Generation

Creating the Parallel NOR Flash Programming File

The ISE tools PROMGen or iMPACT PROM File Formatter creates flash files. These ISE tools take an FPGA bitstream (.bit) and generate a flash file (.mcs) that can be used to program the parallel NOR flash. The options to create the file are described in [Table 5](#).

Table 5: PROMGen Options

PROMGen Option	Description
-p <format>	File format. The most common format is Intel hex (.mcs).
-c FF	Flash fill value.
-s <size>	Flash size in kilobytes (must be a power of 2). If not specified, this defaults to the size of the file(s).
-o <filename>	Output flash file name.
-u <hexaddr> <file[.bit]>	Loads the .bit file from the specified starting address in an upward direction. This option must be specified immediately before the input bitstream file.
-data_width	Specifies the data width of the targeted flash. Bit/byte ordering should be changed between x8 and x16 in the device bitstream depending on the target flash data bus width. The x16 ordering is the only mode supported for BPI Configuration interface synchronous read.
-w	Overwrite existing output file.
-data_file	Specify the starting address, direction, and data files to add into the flash file.
-bpi_dc	Selects a daisy-chain output from the first FPGA. Parallel is the only valid option for the 7 series FPGA BPI Fast Configuration mode.

PROMGen Command Line

The ISE tools PROMGen is easily executed from a command line. An example PROMGen command line is shown below:

```
Promgen -w -p mcs -c FF -bpi_dc parallel -data_width 16 -o flash_file_name
-s 131072 -u 00000000 design.bit
```

This command line generates a parallel NOR file with these options:

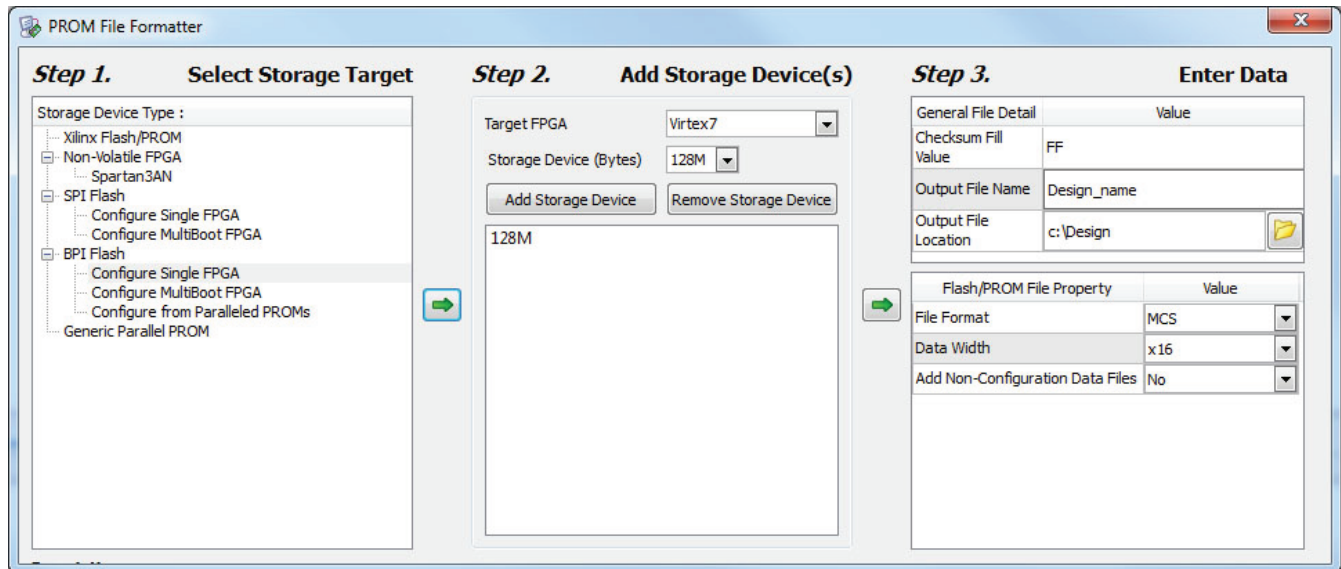
- format = mcs
- data bus = 16
- flash size = 1 Gb (131,072 x 1,024 bytes x 8 bits = 1,073,741,824)
- BPI from address 0
- output file = design.mcs
- fill value = FF
- input bitstream = design.bit

PROM File Formatter GUI Flow Example

Within the iMPACT tool, a flash file can also be generated using these steps:

1. Open iMPACT and double click the **Create PROM File (PROM File Formatter)** iMPACT Flow in the upper left box.
2. A wizard is displayed to guide the user through the next steps ([Figure 8](#)):
 - a. Under “Step 1. Select Storage Target” select **BPI Flash > Configure Single FPGA** and click the green arrow.

- b. Under “Step 2. Add Storage Device(s)” select the target FPGA (e.g., **Virtex-7**) and the storage device (bytes) (**128M** for the 28F00AG18F) and then click **Add Storage Device**. Click the green arrow.
- c. Under “Step 3. Enter Data” select the output file name, output location, and flash properties. Click **OK**.



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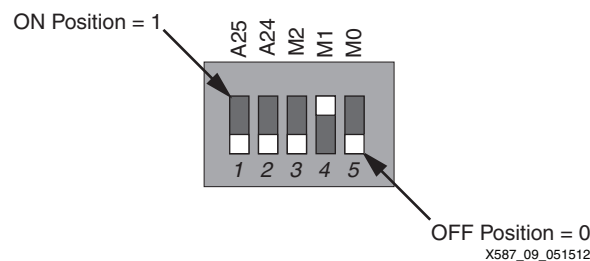
Figure 8: PROM File Formatter Wizard

3. When prompted, click **OK** and add the target user design bitstream (.bit) file. For a single design image, click **No** at the next prompt that is used for multiple designs. Click **OK** to confirm the completion of the design file entry and again at the summary of usage.
4. After successfully running a project, click **Generate File...** to create the flash programming image.

Virtex-7 FPGA VC707 Board Setup for iMPACT Indirect Flash Programming

Refer to the *VC707 Evaluation Board for the Virtex-7 FPGA User Guide* [Ref 7] for detailed information on features and board usage. The basic board setup to program the parallel NOR flash with iMPACT and to properly configure the Virtex-7 FPGA is briefly highlighted below:

1. Ensure the defaults for the Mode pins M[2:0] and the two upper address lines A[25:24] tied to Revision Select RS[1:0] are set as shown in Figure 9.



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Figure 9: VC707 SW1: Mode and RS[1:0] Pin Settings

2. Power on the VC707 board.

3. Connect a USB Type-A to Micro-B cable to the USB JTAG (Digilent) connector on the VC707 board or connect a Xilinx USB II cable to the header.

Cable Options

The iMPACT programming tool uses a Xilinx Platform Cable USB II cable or the Digilent JTAG SMT1 module (on-board) to indirectly program the VC707 parallel NOR flash using the Virtex-7 FPGA JTAG interface. Table 6 lists the cables and the maximum supported frequency for the iMPACT indirect programming solution.

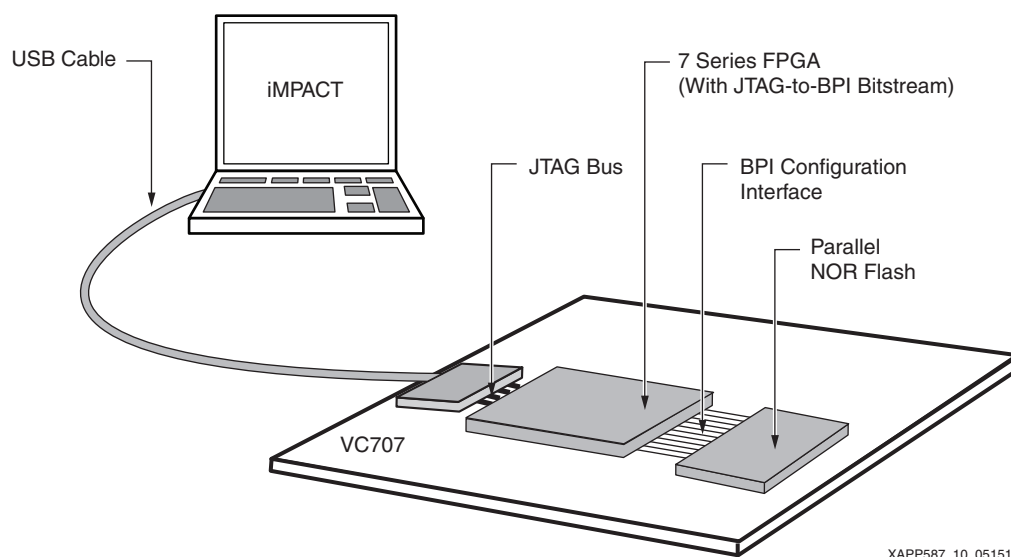
Table 6: VC707 Cable Support

Cables	Interface	BPI Indirect Programming Max Frequency Supported (MHz)
Xilinx Platform Cable USB II	USB	12
Digilent JTAG SMT1 Module (On-Board)	USB	30

Refer to the specific USB cable data sheet for additional information on the JTAG signal connectivity and header details.

iMPACT Indirect Parallel NOR Flash Programming

The iMPACT tool is used to indirectly program the parallel NOR flash. Because parallel NOR flash memory devices do not have a JTAG interface, extra logic is required to serve as a bridge between the iMPACT tool and the flash. For this programming solution, iMPACT configures the FPGA with a bitstream that serves as a bridge between the IEEE 1149.1 (JTAG) bus and the parallel NOR flash bus interface. This solution is referred to as “indirect programming” because the flash is not directly programmed from iMPACT but is programmed through the FPGA. The basic setup of the iMPACT indirect programming solution is shown in Figure 10 for the Virtex-7 FPGA VC707 evaluation board.



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Figure 10: iMPACT Indirect flash Programming Setup

The Virtex-7 XC7VX485T device and the Micron 28F00AG18F parallel NOR flash on the VC707 evaluation board are used to demonstrate the iMPACT programming flow. Before beginning the programming steps, user should have followed the basic board and cable setup details provided in [Virtex-7 FPGA VC707 Board Setup for iMPACT Indirect Flash Programming](#), page 15.

After the board is set up, the on-board flash can be programmed without having to remove it from the board. The 7 series FPGA must first be configured via JTAG with the iMPACT pre-made bitstream that bridges the programming cable to the flash. The existing contents of the FPGA are cleared in order to load the iMPACT pre-made bitstream. The bitstream design 3-states all unused I/Os that are not part of the parallel NOR interface with weak pull-ups. The user should be aware of this step and ensure that this does not have any undesired effect on other devices attached to the FPGA in the application. If there are unused I/Os that should not be pulled up, then external resistor termination would be required. The bitstream design is integrated into the iMPACT programming tool and automatically loaded in the background during an indirect flash operation. The core is for iMPACT runtime use and the source code is not available for design integration. After the bitstream design is loaded, the iMPACT tool issues the command sequence to the flash based on the target device and operation selected.

The iMPACT tool can be run in batch mode or through the GUI. The iMPACT steps to program the flash indirectly are shown in the next section.

iMPACT Batch Example to Indirect Program Parallel NOR Flash

This section shows popular operations and options for running iMPACT in batch mode with a command file.

```
iMPACT -batch design.cmd
```

Sample design.cmd contents:

```
setMode -bs
setCable -port auto
Identify -inferir
identifyMPM
attachflash -position 1 -bpi "28F00AG18F"
assignfiletoattachedflash -position 1 -file "designname.mcs"
Program -p 1 -dataWidth 16 -rs1 25 -rs0 24 -bpionly -e -loadfpga
closeCable
Quit
```

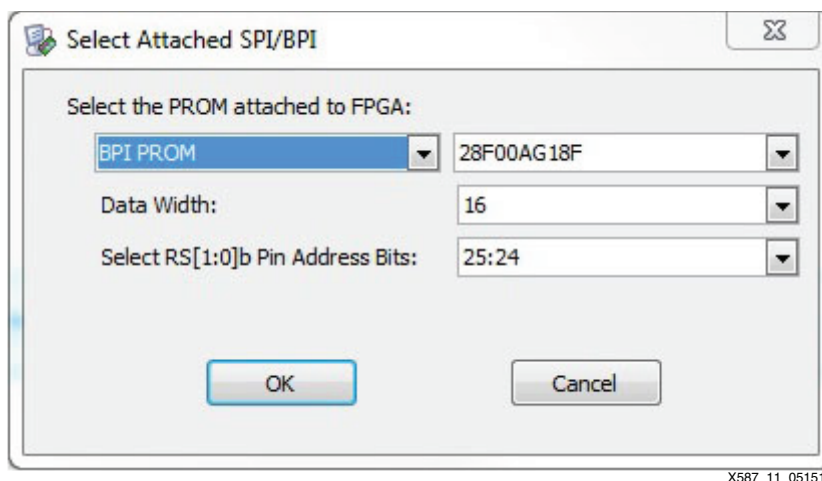
Optional instructions include flash verify and readback commands:

```
BlankCheck -p 1 -dataWidth 16 -rs1 25 -rs0 24 -bpionly -loadfpga
Verify -p 1 -dataWidth 16 -rs1 25 -rs0 24 -bpionly -loadfpga
```

iMPACT Graphical User Interface Example Steps to Program Parallel NOR Flash

1. Open the iMPACT design tool standalone or from within ISE Project Navigator by selecting the **Configure Target Device** process. In iMPACT, double click the **Boundary Scan** selection under the iMPACT Flows in the upper left of the screen.
2. Select the **Initialize Chain** icon to identify the target device on the board.
3. iMPACT prompts the user to assign files. Click **No**.
4. Right click on the **SPI/BPI?** dotted box above the FPGA icon and select **Add SPI/BPI Flash**. When prompted, assign the target flash image.
5. Select **BPI PROM** and **28F00AG18F** for the 1 Gb parallel NOR flash that is on-board the VC707 board. Ensure that the bus data width is set to **16** and that the RS[1:0] selection is **25:24**. Click **OK**.

Note: On the VC707 board, the iMPACT core drives the FPGA RS0 and RS1 signals with this selection because they are the FPGA pins connected to the upper flash address bits A24 and A25, respectively. The RS[1:0] pins are optional and used for external revision control if multiple designs are stored in a single flash. If the RS[1:0] pins are not utilized by a design for revision control, this option during programming should be set to **Not Used**.

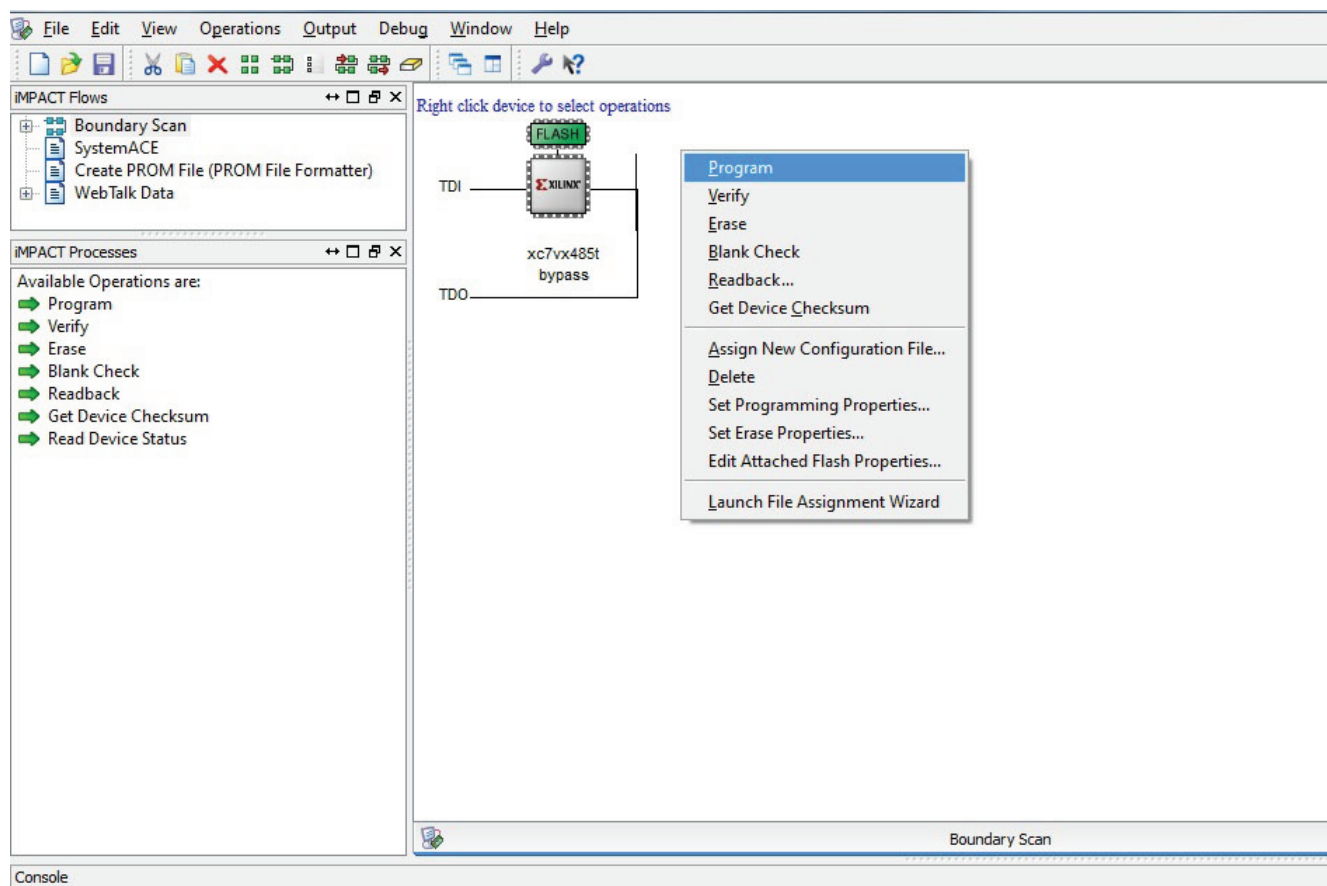


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Figure 11: Flash Target Selection

6. After the parallel NOR flash target has been selected, right click on the flash icon above the FPGA icon to choose the operation to perform on the flash or to select the operation properties. Click on the **Program** operation which by default always performs a design-specific erase, program, and verify.

Note: In a design-specific erase, the flash contents are erased in the address space addressed by the targeted programming file to the nearest flash boundary. To erase the entire flash, the **Set Erase Properties...** selection should be chosen and changed to **Full Chip Erase**.



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Figure 12: Indirect Flash Operation

After issuing the flash operation, the iMPACT tool first downloads the pre-made bitstream and verifies some common parameters for density and family in the common flash interface (CFI). An example iMPACT log display for the CFI query on the VC707 28F00AG18F is shown below:

```
Populating BPI common flash interface ...
Common Flash Interface Information Query completed successfully.
INFO:Cse - Common Flash Interface Information from Device:
INFO:Cse - Verification string: 51 52 59
INFO:Cse - Manufacturer ID:      89
INFO:Cse - Vendor ID:           00
INFO:Cse - Device Code:         1b
Reset Core
Using x16 mode ...
Set Data Width
Using RS Pins [25,24]...
```

IMPACT Indirect Flash Operation Time Estimates

Using iMPACT 14.1 with the VC707 board, some reference sample operation times are provided for compressed and uncompressed XC7VX485T bitstreams using the board's embedded USB cable at a default of 10 MHz (Table 7). These values are for reference only and not guaranteed timing.

Table 7: Reference Sample Flash Operation Times

Image	Times (sec)		
	Erase	Program	Verify
XC7VX485T Sample Compressed 15 Mb	12	90	47
XC7VX485T Uncompressed 157.4 Mb	82	918	466
1 Gb Full Image	103	6612	3345

Effect on the Rest of the System from iMPACT Pre-Made Bitstream

The iMPACT JTAG-to-BPI pre-made bitstream is loaded into the FPGA before performing the target flash operation. The JTAG-to-BPI bitstream sets all unused I/Os to pull-up. If the FPGA I/O is controlling any other devices, pull-ups or pull-downs should be added to disable the device during programming if a pull-up on the I/O will cause contention or undesired behavior.

When the iMPACT bitstream is loaded, it drives the FPGA signals as required by the flash operation. The signals driven include the interface signals A[28:00], D[15:00], FCS_B, FWE_B, FOE_B, ADV_B, and optionally RS[1:0]. The I/O standard used by the design is compatible with the CFGBVS 1.8V or 3.3V selection. When the pre-made bitstream controller is not active, these flash interface signals are 3-stated and should be externally terminated as needed by the user application.

In applications using the RS[1:0] revision select capability, the RS[1:0] pins from the FPGA are tied to the flash upper address signals. Although the design might not have the address pins connected, the core drives both the RS[1:0] and A[28:00] pins. For example, in a 1 Gb flash device, if the upper address lines (A[25:24]) are associated with RS[1:0], both the FPGA A[25:24] signals and the RS[1:0] signals are driven by the core during the flash operation.

Checklist and Debug

This section summarizes the considerations and common debug steps for the BPI Configuration mode and indirect flash programming.

Configuration

- 7 series FPGA Status register data can be read in iMPACT via JTAG. For example, if the DONE and INIT signals are Low, this register captures the specific error conditions that can help identify the type of failure. In addition, the Status register allows the user to verify the Mode pin settings M[2:0] and the bus width detect (for x16, this should be CFG BUS [25:24]=01).
- A bitstream generated for asynchronous mode and the internal configuration clock default can be a useful debug tool if a configuration failure is encountered using the synchronous read. The asynchronous read bitstream test helps to ensure that the interface connectivity on the board is correct and that there is not a maximum clock frequency violation. A new asynchronous read bitstream with the **-g ExtMasterCclk_en:Disable** and **-g BPI_sync_mode:Disable** options would be generated for this test.
- For FPGA designs that do not access the parallel NOR flash, the following is a list of recommendations for maintaining the flash in a safe, low-power condition while the FPGA user design is running:
 - Drive the FPGA FCS_B pin High
 - Drive the FPGA FOE_B pin High
 - Drive the FPGA FWE_B pin High

File Generation

- RTL entry: Ensure that the EMCCLK I/O standard is defined. See [Design Entry Preparation for EMCCLK, page 10](#).
- BitGen usage: Ensure that the Synchronous mode and EMCCLK are enabled with **-g BPI_sync_mode:Type1 -g ExtMasterCclk_en:div-1**. Optionally, for faster configuration and programming times, use the **-g compress** option. Also, ensure that the ConfigRate option does not exceed the maximum supported frequency supported by the target flash and FPGA for asynchronous read mode.
- PROMGen usage: Ensure that the option **-data_width 16** data bus width is selected as described.

iMPACT Indirect Programming

- Erase operation: Flash devices are non-volatile devices and *must* be erased before programming.
- BlankCheck operation: This operation can be used for verifying a prior erase operation.
- Maximum cable speed for flash operations: Ensure that the maximum cable speed for indirect programming is not exceeded. See [Cable Options, page 16](#).
- Revision select pin usage RS[1:0]: These pins support an optional feature. If this feature is used instead of the FPGA upper two address bits for revision control, the option should be selected in iMPACT.

Conclusion

The 7 series BPI Fast Configuration uses the synchronous read and EMCCLK options to significantly decrease the required FPGA configuration time. This mode assists designers with meeting strict power-up requirements of popular standards such as PCIe. The ability to in-system program the parallel NOR flash for this solution provides an additional system value for the 7 series FPGA.

References

This following references provide additional information useful to this application note:

1. [UG470](#), *7 Series FPGA Configuration User Guide*
2. iMPACT Manual
http://www.xilinx.com/support/documentation/sw_manuals/xilinx14_1/isehelp_start.htm#pim_c_overview.htm
3. Micron StrataFlash Embedded Memory
http://www.micron.com/parts/nor-flash/parallel-nor-flash/~media/Documents/Products/Data%20Sheet/NOR%20Flash/Parallel/G18/6702128_256_512_1Gb_65nm_g18.ashx
4. [DS181](#), *Artix-7 FPGAs Data Sheet: DC and Switching Characteristics*
5. [DS182](#), *Kintex-7 FPGAs Data Sheet: DC and Switching Characteristics*
6. [DS183](#), *Virtex-7 FPGAs Data Sheet: DC and Switching Characteristics*
7. [UG885](#), *VC707 Evaluation Board for the Virtex-7 FPGA User Guide*
8. Virtex-7 FPGA VC707 Evaluation Kit
<http://www.xilinx.com/products/boards-and-kits/EK-V7-VC707-G.htm>
9. ISE software manuals
http://www.xilinx.com/support/software_manuals.htm
10. iMPACT Flash Device Support Table
http://www.xilinx.com/cgi-bin/docs/rdoc?v=latest_ise;d=isehelp_start.htm;a=pim_c_introduction_indirect_programming.htm

Revision History

The following table shows the revision history for this document.

Date	Version	Description of Revisions
05/25/12	1.0	Initial Xilinx release.
05/31/12	1.0.1	Updated PDF document properties.

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